EXP 27

```
clc;
clear;
close all;

% Processing Gain and PN Sequence Generation
% for Spread Spectrum Communication

% Step 1: Define Parameters
N = 7;                  % PN sequence length
dataBits = [1 0 1 1 0 1 0];
Rb = 1000;              % Data bit rate (bps)
Rc = 7000;              % Chip rate (chips/sec)

% Step 2: Generate PN Sequence using 3-bit LFSR
% Feedback polynomial: x^3 + x + 1
register = [1 1 1];
PN = zeros(1,N);

for i = 1:N
    PN(i) = register(3);

    % XOR feedback
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    feedback = xor(register(3), register(1));

    % Shift register
    register = [feedback register(1:2)];
end

% Step 3: Display PN Sequence
disp('Generated PN Sequence:');
disp(PN);

% Step 4: Calculate Processing Gain
PG = Rc/Rb;
PG_dB = 10*log10(PG);

fprintf('\nData Rate = %d bps\n', Rb);
fprintf('Chip Rate = %d chips/sec\n', Rc);
fprintf('Processing Gain = %.2f\n', PG);
fprintf('Processing Gain in dB = %.2f dB\n', PG_dB);

% Step 5: Convert Data Bits and PN Bits to Bipolar Form
dataBipolar = 2*dataBits - 1;
PNbipolar = 2*PN - 1;

% Step 6: Spread Data
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spreadSignal = zeros(1,length(dataBits)*N);

for i = 1:length(dataBits)
    idx = (i-1)*N + (1:N);
    spreadSignal(idx) = dataBipolar(i)*PNbipolar;
end

% Step 7: Despread the Signal
recoveredBits = zeros(1,length(dataBits));

for i = 1:length(dataBits)
    idx = (i-1)*N + (1:N);

    % Correlation with original PN sequence
    correlation = sum(spreadSignal(idx).*PNbipolar);

    recoveredBits(i) = correlation > 0;
end

% Step 8: Display Results
disp('Original Data Bits:');
disp(dataBits);

disp('Spread Signal:');
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```
disp(spreadSignal);

disp('Recovered Data Bits:');
disp(recoveredBits);

% Step 9: Plot PN Sequence
figure;

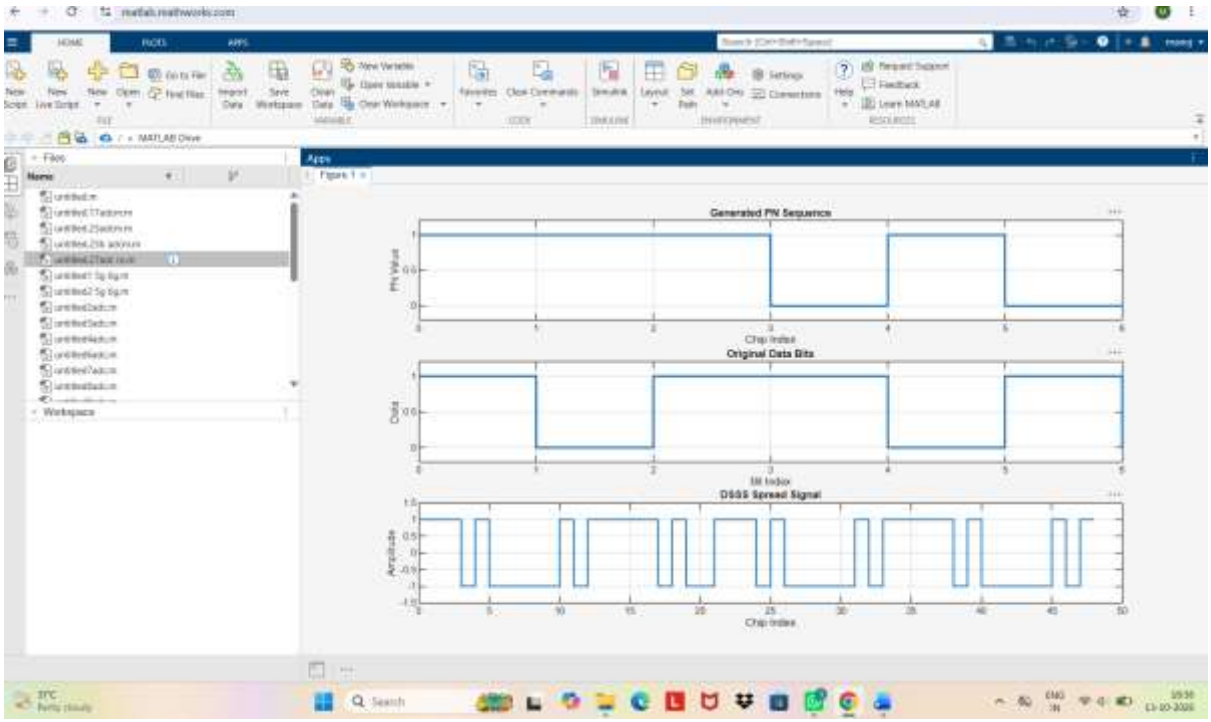
subplot(3,1,1);
stairs(0:N-1,PN,'LineWidth',2);
grid on;
ylim([-0.2 1.2]);
xlabel('Chip Index');
ylabel('PN Value');
title('Generated PN Sequence');

% Step 10: Plot Original Data
subplot(3,1,2);
stairs(0:length(dataBits)-1,dataBits,'LineWidth',2);
grid on;
ylim([-0.2 1.2]);
xlabel('Bit Index');
ylabel('Data');
title('Original Data Bits');
```

```
% Step 11: Plot Spread Signal
subplot(3,1,3);
stairs(0:length(spreadSignal)-1,spreadSignal,'LineWidth',1.5);
grid on;
ylim([-1.5 1.5]);
xlabel('Chip Index');
ylabel('Amplitude');
title('DSSS Spread Signal');

% Step 12: Verify Recovery
if isequal(dataBits,recoveredBits)
    disp('SUCCESS: Original data recovered correctly.');
else
    disp('ERROR: Data recovery failed.');
```

end


Generated PN Sequence
PN Value
Chip Index
Original Data Bits
Data
Bit Index
DSSS Spread Signal
Amplitude
Chip Index